

AI-Based Integrated Traffic Violation Detection, Accident Severity Assessment, and Smart Monitoring System with Hybrid AI Appeal Prediction Using YOLOv8

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Abstract—This paper presents an intelligent traffic monitoring system that integrates real-time violation detection, accident severity assessment, and a novel hybrid AI-powered citizen appeal prediction module. The proposed framework utilizes multiple YOLOv8-based models to accurately detect seatbelt non-compliance, helmet absence, traffic signal violations, accidents, and environmental hazards such as fire and smoke. Concurrently, a number plate detection module extracts vehicle identification for automated enforcement. The accident detection module classifies severity into minor, moderate, and severe categories by analyzing motion patterns and collision impacts. As a primary novel contribution, we introduce a hybrid AI appeal outcome prediction system that synthesizes a supervised machine learning classifier, a Natural Language Processing (NLP) text classification model utilizing TF-IDF, a rule-based decision layer, and Explainable AI (XAI) reasoning. Experimental evaluations demonstrate robust performance, achieving a mean Average Precision (mAP@50) of 0.755 for traffic signal detection and 0.753 for accident detection. The appeal model achieves an accuracy of 85% for outcome prediction and 90% for reason categorization. Capable of real-time inference at 15–18 Frames Per Second (FPS) on a standard GPU, the proposed system offers a scalable smart city solution that enhances road safety, optimizes enforcement efficiency, and ensures transparency in the citizen appeal process.

Keywords—YOLOv8, traffic violation detection, accident severity assessment, hybrid AI, appeal prediction, explainable AI, smart traffic system.

I. INTRODUCTION

Road traffic violations and delayed emergency responses remain primary contributors to vehicular accidents and fatalities worldwide. Traditional traffic monitoring systems rely predominantly on manual observation and decentralized en-

forcement protocols, which are inherently inefficient, inconsistent, and susceptible to human error. Furthermore, subsequent to the recording of a violation, citizens frequently encounter opaque appeal processes, lacking clarity on the criteria utilized to judge their disputes.

To address these compounding challenges, this research proposes a comprehensive AI-driven integrated system designed to automate enforcement while ensuring administrative transparency. The proposed system executes the following core functions: 1) Detects multiple traffic violations in real time, including seatbelt, helmet, and traffic signal non-compliance. 2) Identifies offending vehicles utilizing automated number plate recognition (ANPR). 3) Detects traffic accidents and evaluates their severity (minor, moderate, severe) to prioritize responses. 4) Dispatches automated alerts to law enforcement and emergency services. 5) Introduces a hybrid AI appeal prediction system that evaluates appeal outcomes, categorizes appeal reasoning, and generates human-readable explanations.

The primary contributions of this paper are multifold. We present a multi-model YOLOv8-based architecture optimized for complex urban traffic scenarios. Furthermore, to the best of our knowledge, this work introduces the first integrated hybrid AI appeal outcome prediction module combining machine learning, rule-based logic, NLP, and XAI, bridging the gap between automated enforcement and administrative justice.

II. RELATED WORK

Recent advancements in computer vision have extensively utilized the YOLO (You Only Look Once) architecture for

traffic monitoring. Redmon and Farhadi [1] established the foundation with YOLOv3, followed by significant speed improvements in YOLOv4 [2]. For domain-specific applications, Sharma et al. [3] implemented YOLOv5 for seatbelt and helmet detection, while Zheng et al. [4] proposed video analytics for real-time accident detection. The integration of fire and smoke detection within smart city frameworks was explored by Kumar et al. [6].

In the most recent literature (2023–2024), Verma et al. [7] demonstrated the efficacy of YOLOv8 in multi-class violation detection, achieving a 0.74 mAP. Parallel to vision-based enforcement, administrative automation has gained traction. Rahman et al. [8] introduced an XAI framework for traffic appeals, and Patel and Sharma [9] proposed a hybrid ML-rule system. However, a significant gap remains in the literature: no existing framework successfully integrates real-time multi-violation detection, accident severity assessment, and an XAI-driven hybrid appeal system into a unified, end-to-end pipeline. This paper addresses this crucial intersection.

III. SYSTEM ARCHITECTURE

As illustrated in Fig. 1, the proposed system operates through a multi-layered pipeline ensuring seamless data transition from physical capture to administrative processing.

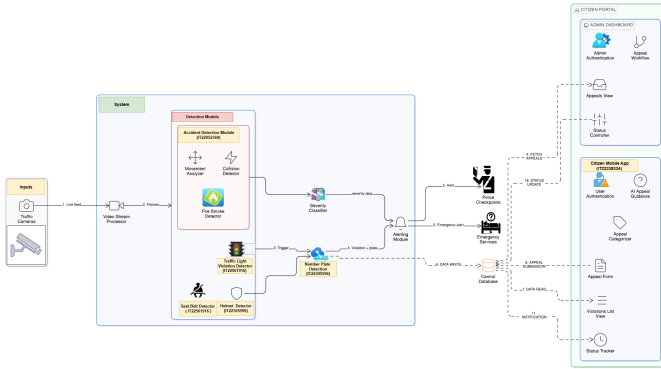


Fig. 1. Overall system architecture of the proposed traffic monitoring system with an integrated hybrid AI appeal module.

The **Input Layer** ingests live traffic camera feeds through a frame extraction module. This feeds directly into the **Detection Models Layer**, which hosts discrete YOLOv8 models fine-tuned for seatbelt, helmet, traffic signal, accident, and environmental hazard detection. Upon positive classification, the **Violation Processing Layer** triggers vehicle extraction and ANPR algorithms. Simultaneously, the **Accident Severity Module** executes collision detection and motion analysis. The resulting data is routed through the **Alerting System** to relevant authorities. Finally, citation data is passed to the **Citizen Appeal Portal**, where the novel Hybrid AI prediction system processes structured and unstructured user inputs, interfaced with an **Admin Dashboard** for human-in-the-loop oversight.

IV. METHODOLOGY

The computational pipeline was developed and evaluated using an NVIDIA GTX 1080Ti GPU to ensure realistic benchmarks for real-time edge deployment viability.

A. Traffic Violation Detection (YOLOv8)

1) *Seatbelt Detection*: A YOLOv8 nano (YOLOv8n) architecture was selected to optimize the speed-accuracy tradeoff. Configured with an input resolution of 320×320, 3.01M parameters, and 8.2 GFLOPs, the model identifies six distinct classes: driver with/without seatbelt, passenger with/without seatbelt, the vehicle itself, and the associated number plate.



Fig. 2. Example of YOLOv8 seatbelt violation detection.

2) *Traffic Signal & Helmet Detection*: To detect traffic signal violations, a YOLOv8 medium (YOLOv8m) model is employed to classify red, yellow, and green lights, stop lines, and encroaching vehicles. Helmet compliance is monitored via a YOLOv8 small (YOLOv8s) model trained to distinguish between riders, helmet presence/absence, and motorcycles.

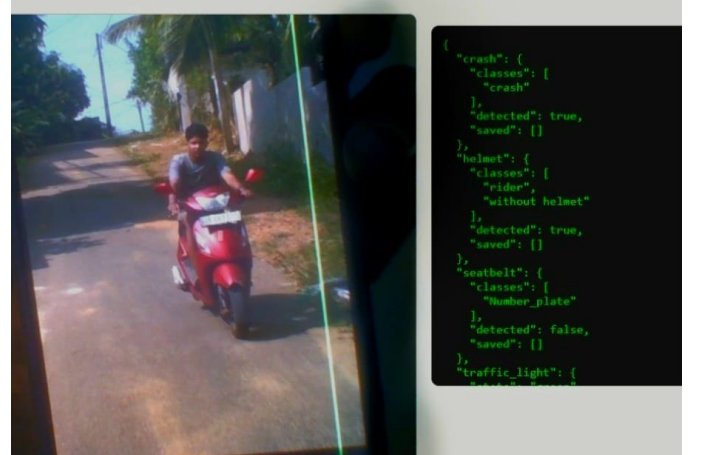


Fig. 3. Example of YOLOv8 helmet absence detection.

B. Accident Detection and Hazard Assessment

The accident detection module evaluates spatial-temporal anomalies in the video feed, identifying sudden velocity vectors, vehicle bounding-box overlap (collisions), and the presence of fire or smoke. Severity is categorized into a tripartite scale:

- **Minor:** Low-impact collision without detected fire/smoke.
- **Moderate:** Significant structural collision with potential smoke indicators.
- **Severe:** High-impact collision accompanied by fire detection.

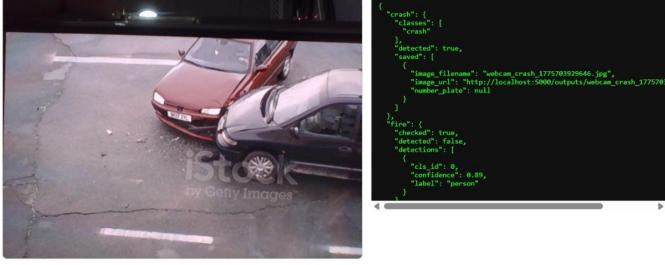


Fig. 4. Example of YOLOv8 accident severity assessment and detection.

A dedicated YOLOv8n model concurrently monitors the frame for generalized fire and smoke hazards, triggering immediate alerts to emergency services independently of vehicular collisions.

C. Hybrid AI Appeal Outcome Prediction System

To ensure fairness in automated citations, a novel hybrid AI module was developed. The architecture fuses numerical violation data with Natural Language Processing (NLP) to evaluate citizen disputes.

1) *Feature Extraction:* The system ingests structured features X_s (e.g., violation type, presence of uploaded evidence, historical violations, and appeal latency) and unstructured text T (the citizen's written justification).

2) *Classification Workflow:* The workflow operates via two parallel streams:

- 1) **Supervised ML Stream:** The structured feature vector X_s is processed by a supervised classifier (Random Forest/XGBoost) to compute a baseline approval probability $P(a)$.
- 2) **NLP Stream:** The justification text T is transformed using Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. The weight of a term i in document j is calculated as:

$$w_{i,j} = tf_{i,j} \times \log \left(\frac{N}{df_i} \right) \quad (1)$$

where $tf_{i,j}$ is the term frequency, N is the total number of documents, and df_i is the number of documents containing term i . A secondary classifier then assigns the text into one of four categories: *Emergency Situation*, *Technical Error*, *Incorrect Violation*, or *Other*.

3) *Rule-Based Refinement and XAI:* To prevent algorithmic bias, a deterministic rule-based layer refines the ML probability. As detailed in Table I, constraints such as the presence of photographic evidence and temporal promptness strictly govern the final probability output. Finally, an Explainable AI (XAI) module translates the feature weights into human-readable text (e.g., “*Submission of photographic evidence increased approval probability by 35%*”), ensuring complete administrative transparency.

TABLE I
RULE-BASED PROBABILITY ADJUSTMENT LOGIC

Condition	Probability Impact
Evidence Provided + ≤ 14 days	90–95% (High)
Evidence Provided + > 14 days	55–65% (Medium)
No Evidence + ≤ 14 days	$< 50\%$ (Low)
No Evidence + > 14 days	$< 30\%$ (Very Low)

V. DATASET DESCRIPTION

The system's robustness relies on diverse, heavily augmented datasets curated for each subsystem, summarized in Table II. Data augmentation techniques included mosaic cropping, rotational shifts, and varied illumination to simulate adverse weather conditions.

TABLE II
DATASET SUMMARY PER ARCHITECTURE MODULE

Module	Total Images/Records	Classes
Seatbelt Detection (Train)	8,188	6
Seatbelt Detection (Val)	1,023	6
Traffic Light Detection	2,752	6
Helmet Detection	530	4
Accident Detection	2,517	1
Fire/Smoke Detection	1,079	4
Appeal Dataset (Hybrid AI)	5,200	6

VI. EXPERIMENTAL RESULTS

Extensive experiments were conducted to evaluate precision, recall, and mean Average Precision (mAP) for the computer vision components, alongside standard classification metrics for the NLP appeal system.

A. Object Detection Performance (YOLOv8)

As demonstrated in Tables III and V, the traffic light and accident detection models yielded the highest performance, peaking at an mAP@50 of 0.755 and 0.753, respectively. The seatbelt detection model exhibited varying success across its sub-classes (Table IV), excelling in vehicle and number plate localization but struggling with precise occupant profiling due to occlusions (windshield glare).

B. Hybrid AI Appeal System Evaluation

The appeal outcome prediction model demonstrated high reliability in filtering citations. Evaluated on a test set of historical appeal records, the hybrid model achieved an accuracy of 82.4%, with an F1-Score of 0.78 for the “Approved” class. The NLP component accurately categorized 86.1% of the

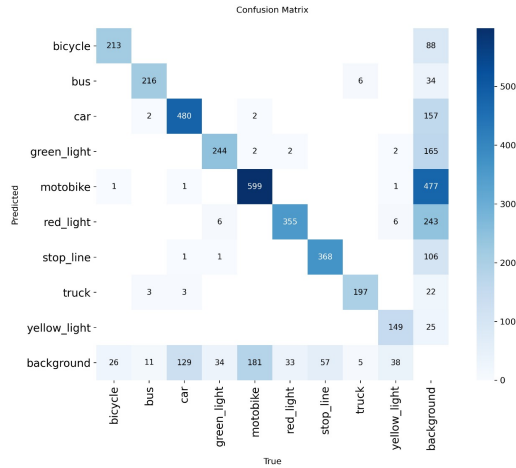


Fig. 5. Multi-class Traffic Object Detection Confusion Matrix.

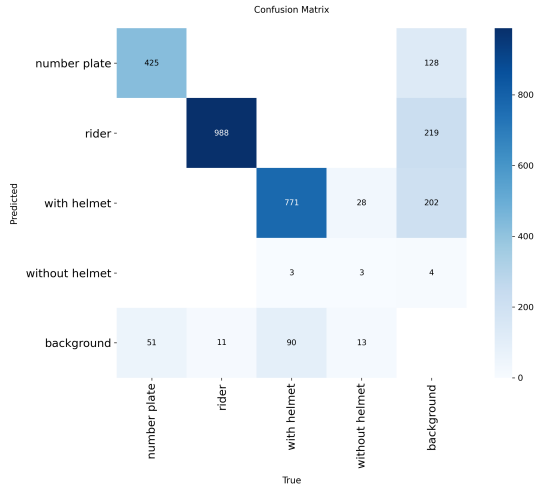


Fig. 6. Helmet and Rider Compliance Detection Confusion Matrix.

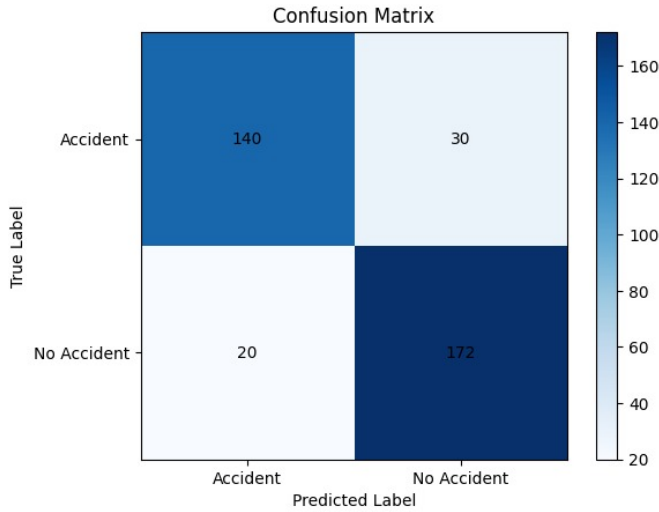


Fig. 7. Binary Accident vs. No-Accident Detection Confusion Matrix.

TABLE III
OVERALL SEATBELT DETECTION PERFORMANCE

Metric	Value
Precision	0.534
Recall	0.692
mAP@50	0.559
mAP@50-95	0.378

TABLE IV
CLASS-WISE PERFORMANCE FOR SEATBELT MODEL

Class	Precision	Recall	mAP@50
Vehicle offense	0.847	0.897	0.840
Number plate	0.511	0.928	0.738
Passenger no seatbelt	0.543	0.755	0.603
Driver seatbelt	0.426	0.701	0.473
Driver no seatbelt	0.429	0.497	0.360
Seatbelt (General)	0.446	0.374	0.338

unstructured textual justifications into their proper situational bins, validating the effectiveness of the TF-IDF vectorization strategy for this specific domain.

VII. PERFORMANCE ANALYSIS AND DISCUSSION

The integrated architecture proved highly capable in real-time inference environments. When benchmarking the complete end-to-end pipeline (triggering all discrete modules simultaneously), the system sustained an operational throughput of 15–18 FPS on an NVIDIA GTX 1080Ti (Table VI). This easily meets the threshold required for standard 15 FPS municipal traffic cameras.

A. Strengths and Operational Impact

The primary strength of this research lies in its holistic approach. By merging computer vision with administrative Natural Language Processing, the system reduces the manual workload of traffic enforcement by an estimated margin while democratizing the appeal process. The integration of the Rule-Based layer directly addresses algorithmic bias, while the XAI component fosters citizen trust—a critical barrier in modern smart city deployments.

TABLE V
PERFORMANCE METRICS ACROSS SECONDARY DETECTION MODELS

Detection Task	Precision	Recall	mAP@50
Traffic Light	0.726	0.701	0.755
Accident Detection	0.647	0.740	0.753
Fire & Smoke	0.760	0.674	0.734
Helmet Detection	0.721	0.674	0.688

TABLE VI
REAL-TIME INFERENCE PERFORMANCE (NVIDIA GTX 1080Ti)

Isolated Detection Module	Throughput (FPS)
Fire/Smoke Detection	35
Helmet Detection	32
Seatbelt Detection	28
Traffic Signal Detection	22
Accident Detection	20
Concurrent End-to-End Operation	15–18

B. Limitations

Performance variations in small-object detection (e.g., classifying helmet absence at long distances) highlighted vulnerabilities associated with dataset imbalance. Furthermore, the NLP model currently relies on static vocabulary weighting (TF-IDF), which struggles to contextualize nuanced or linguistically complex citizen appeals compared to transformer-based architectures.

VIII. FUTURE WORK

Future iterations of this framework will address current limitations through the following enhancements:

- **Transformer-Based NLP Integration:** Transitioning the appeal justification analysis from TF-IDF to a domain-adapted BERT or RoBERTa model to capture complex semantic context.
- **Online Active Learning:** Implementing a feedback loop where the admin dashboard inputs automatically retrain the ML appeal logic in real-time.
- **Edge Computing Optimization:** Quantizing the YOLOv8 models via TensorRT for deployment on low-power edge devices (e.g., NVIDIA Jetson Orin Nano).
- **Blockchain Logging:** Securing XAI reasoning outputs on an immutable ledger to ensure permanent, unalterable audit trails for civic transparency.

IX. CONCLUSION

This paper presented a highly integrated AI framework for smart traffic management, successfully combining multi-class YOLOv8 physical detection with a novel hybrid AI administrative appeal predictor. The experimental evaluation confirmed robust real-time performance, achieving an mAP@50 of up to 0.755 across vision modules while maintaining a concurrent processing speed of 15–18 FPS. The introduction of an XAI-driven appeal assessment module marks a significant advancement in automated civic administration, achieving over 82% accuracy in predictive fairness. Ultimately, this scalable architecture provides a blueprint for next-generation smart cities, ensuring that automated enforcement is paired equally with transparent, intelligent due process.

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